Samuel Shilling

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Profile

Quantitative researcher with advanced experience in developing Python-based data processing pipelines, modelling complex datasets using Bayesian inference, and building simulations to forecast future results. Proven ability process complex datasets, identify statistical relationships, and quantify uncertainties. Published first-author research in leading peer-reviewed journals, presented at five research conferences, and research experience at NASA Goddard Space Flight Center. Seeking to apply rigorous mathematical and computational techniques to quantitative research, financial markets and commodity trading.

Skills

- **Programming and tools:** Python, Linux, Git, NumPy, pandas, SciPy, Matplotlib, Ollama (local LLM integration), APScheduler. Automated workflows and data pipelines.
- **Statistical analysis:** Data cleaning and visualisation, MCMC simulations, Spearman's rank and partial correlation tests, bootstrapping, Kolmogorov–Smirnov tests, bias tests, forecasting future trends.
- **Data modelling:** Bayesian inference, nested sampling, posterior estimation, evidence-based model comparison, uncertainty quantification, non-linear least squares, linear regression, model extrapolation.

Experience

Doctoral Researcher, Astrophysics **Oct 2023 – Feb 2027 (expected)**
Lancaster University, UK

- Discovered a statistically significant correlation in a new data domain, ruled out sample and selection biases, and quantified its properties.
- Built a Python pipeline using CPU multiprocessing to model a large sample of complex astrophysical sources using Bayesian inference for model comparison and parameter estimation. Developed a simulation to test the inference.
- Presented research at five conferences (including three major national conferences) to expert and non-expert audiences; assisted supervising four MSc Astrophysics student research projects and teaching undergraduate physics courses; chaired the University Astrophysics journal club in rotation; led teams on NASA and ESA proposals.
- Developed a Python application (available on GitHub) that fully automates journal club role allocation indefinitely. This includes a locally hosted natural-language (Ollama) interface, so members can check and modify their roles in plain English, and a scheduled daily workflow.

Research Assistant, Astrophysics **Jul 2024 – Dec 2024**
NASA Goddard Space Flight Center, MD, USA

- Built a Python pipeline to collate and process 20 years of archival GRB data observed with NASA's *Swift* satellite.
- Discovered data quality (rather than the physics) was a major driver of uncertainty in previous published results. Built a simulation to predict the results in future datasets.
- Analysed newly detected transient astrophysical sources from NASA's *Swift* satellite, in real time (within minutes to hours) producing 37 community bulletin reports.

Education

PhD, Astrophysics **Oct 2023 – Feb 2027 (expected)**
Lancaster University, UK

- Thesis: "Statistical Analyses and Bayesian Inference in Populations of Gamma-Ray Burst Afterglows".

MPhys, Physics and Astrophysics **Sep 2018 – Jun 2023**
University of Sheffield, UK

- Grade: Class one honours (78% final year, 80% dissertation, 77% literature review).

First-author peer-reviewed publications

- Shilling et al. (in prep.) *Inferring GRB jet structures, physical parameters, and geometric properties from a large sample of X-ray afterglows.* In preparation (est. 2027).
- Shilling et al. (2026) *The intrinsic luminosity–decay correlation in subsamples of GRB X-ray afterglows.* MNRAS, 552, stag1589
- Shilling et al. (2025) *Evidence for an intrinsic luminosity–decay correlation in GRB radio afterglows.* MNRAS, 542, 2421